

UDAYKIRAN KALYANAPU

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SUMMARY: Data Engineer with 3+ years of experience designing and optimizing cloud native data workflows on AWS and Microsoft Fabric. Proven expertise in building scalable ETL pipelines, serverless architectures, and analytics solutions using Lambda, S3, Glue, Redshift, and Fabric Lakehouse. Skilled in Terraform, CI CD pipelines, Python, and SQL with a strong focus on data reliability, performance optimization, and cost efficient cloud

EXPERIENCE:

CAPGEMINI *Senior Software Engineer (October 2022 – December 2023) + Software Intern (From Feb 2022)*

- Built end to end **ETL** workflows integrating multiple data sources using **Python, pandas, SQL** and **PySpark**, ensuring data reliability through validation and transformation logic.
- Implemented **infrastructure as code** using **Terraform** to provision **AWS** resources, ensuring repeatable environments and reducing setup time by 30%.
- Developed dashboards and analytical reports using **Power BI** to communicate performance to business stakeholders.
- Worked in **Agile** teams, partnering with **QA**, analytics, and product teams to deliver customer focused cloud solutions.
- Recognized with the **New Generation Talent Hackathon Winner Award (2023)** for innovative contributions in automation and full-stack integration of project **Flight Ticket Booking System**.

PROJECTS:

Intelligent Document Query System | AWS, Amazon Bedrock, Python, S3, Aurora Serverless, Terraform

- Built a **serverless document ingestion** and querying platform on **AWS** enabling natural language search across **100+ unstructured documents** stored in S3 using foundation models via Amazon Bedrock.
- Automated **document ingestion** and **preprocessing pipelines** using **Python** and **AWS SDKs**, reducing manual handling.
- Implemented a scalable metadata and indexing layer using **Aurora Serverless** to store document attributes, **embeddings** references, and query mappings within a secured **VPC**.
- Reduced document retrieval time from minutes to seconds while improving system scalability, reliability, and operational efficiency through fully managed **AWS** services.
- Provisioned complete cloud infrastructure using **Terraform** including **S3, Aurora Serverless, IAM roles**, and networking, reducing environment setup time by **50%**.

Contoso School Parking | Microsoft Power Apps, Microsoft Fabric, Power Automate, Dataverse, Azure Cloud

- Stored and managed application data in **Microsoft Dataverse** and integrated it with **Microsoft Fabric OneLake** for centralized data storage and analytics.
- Built automated workflows using **Power Automate** to trigger data ingestion into **Fabric Lakehouse**, enabling downstream analytics and reporting using **Medallion architecture** of Bronze Silver Gold layering.
- Developed Fabric notebook based transformations using **PySpark** and **Microsoft T-SQL** to clean and aggregate application data, implementing dimensional modeling with **Star Schema** for optimized analytical queries.
- Enabled near real time insights by connecting Fabric datasets to **Power BI** dashboards, reducing manual reporting effort.
- Implemented role based access control and cloud security best practices across **Power Apps, Fabric**, and **Dataverse**.

CERTIFICATIONS:

- AWS Certified Solutions Architect – Associate (SAA C03)
- Microsoft Fabric Data Engineer Associate (DP 700)
- Google Associate Cloud Engineer (ACE)
- Oracle Cloud Infrastructure 2025 Generative AI Professional (OCI 2025 GA)
- Microsoft Power Apps Developer Associate (PL 900)

TECHNICAL SKILLS:

- **Cloud Platforms:** AWS (Lambda, S3, EC2, Glue, Redshift, DynamoDB, Textract, API Gateway, CloudWatch), Microsoft Fabric (OneLake, Lakehouse, Notebooks, Data Pipelines), Azure.
- **Programming and Query Languages:** Python, Java, SQL, T-SQL, NoSQL, JavaScript, TypeScript, Kotlin, Bash, PowerShell.
- **Other Technologies and Tools:** Postman, Tableau, Power BI, Power Automate, Microsoft Power Apps, Microsoft Dataverse, Jenkins, Git, GitHub Actions, REST APIs, Swagger, Maven, Gradle.

EDUCATION:

The George Washington University (CGPA: 3.94/4.0)

Washington, DC

Master of Science, Computer Science

January 2024 - December 2025

- Have worked as a **Graduate Teaching Assistant for Computer System Architecture** (Aug 2024 – Dec 2025)
- Coursework: Advanced Software Paradigms, Design Analysis and Algorithms, Computer System and Architecture, Data Structures, Algorithms, Object Oriented Design, System Design, Distributed Systems, Query Optimization, SDLC